Abhishek Nigam

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Summary

Frontend-Focused Software Engineer with 2.4+ years of experience building scalable, high-performance web applications using React.js and Next.js. Experienced in server-side rendering (SSR), performance optimization, and state management, improving Core Web Vitals for production systems serving 200K+ users. Skilled in backend development with Node.js and REST APIs, contributing to end-to-end system design and reliability. Filed a U.S. patent for AI-driven optimization, leveraging data-driven decision to improve infrastructure efficiency.

Skills

Languages: JavaScript, TypeScript, C++, Java, SQL

Frontend: React.js, Next.js, React Query, Redux, Tailwind CSS, Webpack, SEO Optimization

Backend & Databases: Node.js, REST APIs, API Integration, MongoDB, MySQL

AI/ML: Generative AI, LLMs, RAG, Prompt Engineering, Ollama

Cloud & Tools: Git, GitHub, IBM Cloud, Chrome DevTools, Cypress, Vite, Jenkins, CI/CD Pipelines, Jira

Experience

Software Engineer — **Jan 2024 – Present**
IBM — Bengaluru

- Scaled platform throughput by enabling **4× faster VM restart recovery**, optimizing **Redux state management** and async API handling, supporting onboarding of **200K+ customers.**
- Implemented **Redis-backed caching with React Query** using stale-while-revalidate strategy, optimizing data-fetching and **client-side rendering (CSR)**, reducing **UI latency by 50% (4s → 2s)** and improving frontend performance.
- Designed **RBAC-based** conditional rendering with feature-level access, reducing user-facing **downtime by 40%** on Power11 systems.
- Reduced disk-bound latency using **in-memory caching with lazy loading and pagination**, improving load time **by 35%** and **optimizing Core Web Vitals.**
- Automated UI workflows using **Cypress** with API validation and CI/CD**,** achieving end-to-end coverage and **reducing defects by 30%.**
- Diagnosed and resolved **20+ production** issues using **DevTools and react Profiler**, reducing incidents and improving system stability.

AI/ML Intern — **Jul 2023 – Dec 2023**
IIIT Delhi — New Delhi

- Built an **AI-powered recipe generation** system using **GPT-2,** leveraging **prompt engineering** for cuisine-based personalization. (Demo)
- Fine-tuned **LLM** outputs using **input-conditioned prompts** (ingredients + cuisines) and optimized generation parameters to reduce repetition, **improving relevance by 60%** and **reducing manual effort by 70%.**

Projects

Collab Canvas | React.js, TypeScript, Node.js, Context APIs, WebSockets

- Proposed a **real-time collaborative whiteboard** supporting **10+ concurrent** users with **room-based session isolation.**
- Designed an **event-driven WebSocket** architecture **handling 1000+ events/session**, ensuring low-latency synchronization and consistent state across clients. (Live | GitHub)

Infra Optimizer | Next.js, JavaScript, MongoDB, REST APIs, AI-assisted systems

- Developed an **AI-driven dashboard** analyzing **VM usage metrics** to detect **underutilized resources** and provide infrastructure insights.
- Integrated **Slack alerting and CI/CD workflows** into an AI-driven system, enabling automated decision-making, reducing response time by **10 minutes**, and improving cloud resource visibility.

Education

Indraprastha Institute of Information Technology, Delhi — **Jul 2022 – Jun 2024**
M.Tech in Computer Science and Engineering — CGPA: 8.62

Reva University, Bengaluru — **Jul 2017 – Jun 2021**
B.Tech in Computer Science and Engineering — CGPA: 9.59

Achievements

- Filed a **U.S. patent** on "**AI-Driven Adaptive VM Placement"**, proposing **30-40% cloud resource optimization**. (GitHub)
- Certified in **Microsoft Azure Data Fundamentals** (DP-900) and **Google Analytics**. (Link)
- Secured **AIR 2075** in GATE CSE 2022 and cleared **GATE CSE in 2021 and 2022.** (Link)